The following excerpt is taken from the "**Completed Work**" section of my final-year project progress report,

*Investigating the Relationship Between*
*MRI Synthesis Quality Metrics*
*and Downstream Segmentation Performance*

Earlie sections of the report established the researh context by reviewing prior work on MRI synthesis, image-quality evaluation, and brain-tumour segmentation, and by identifying limitations in existing evaluation approaches. To avoid repetition and focus on original contributions, this excerpt presents the experimental methodology, analysis, and findings developed in the project.

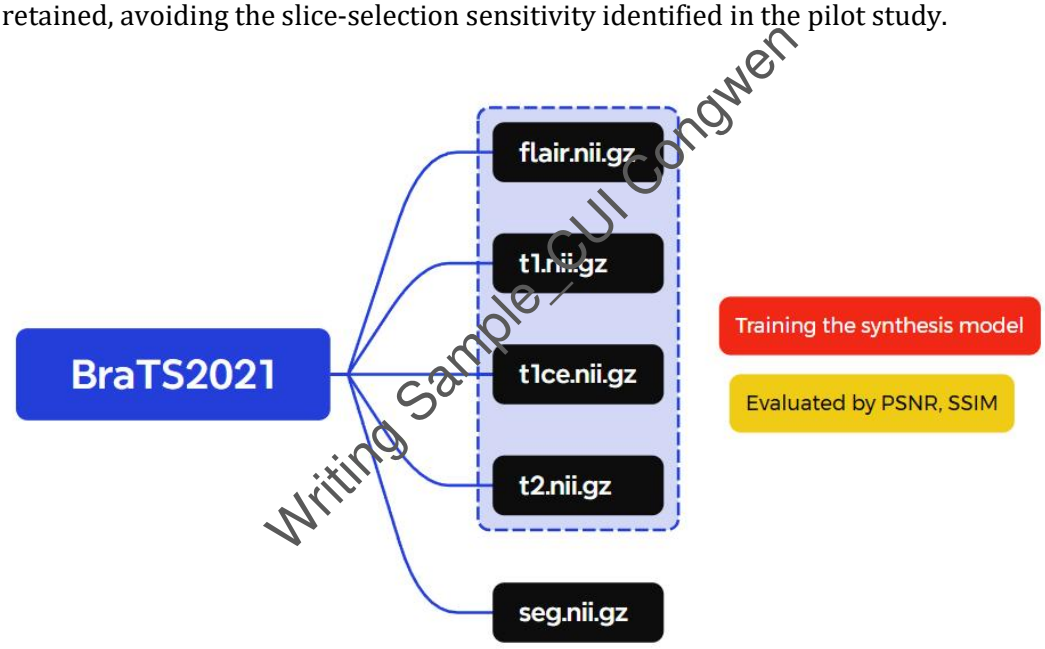
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# 3 Completed Work

This chapter reports the work completed by 11 November 2026. It covers formal dataset preparation, synthesis and segmentation model training, checkpoint-based downstream evaluation, and checkpoint-level and patient-level analyses of the relationship between image similarity and segmentation performance.

## 3.1 Dataset and Experimental Design

The formal experiment used 1,251 patients from the BraTS 2021 training dataset. Patients were separated before model training into 1,001 training, 125 validation and 125 held-out test cases. The training set was used to estimate model parameters, the validation set was used to select checkpoints, and the test set was reserved for the final relationship analysis. Each patient contributed co-registered T1, T2, FLAIR, T1ce and segmentation volumes. All 155 axial slices were retained, avoiding the slice-selection sensitivity identified in the pilot study.



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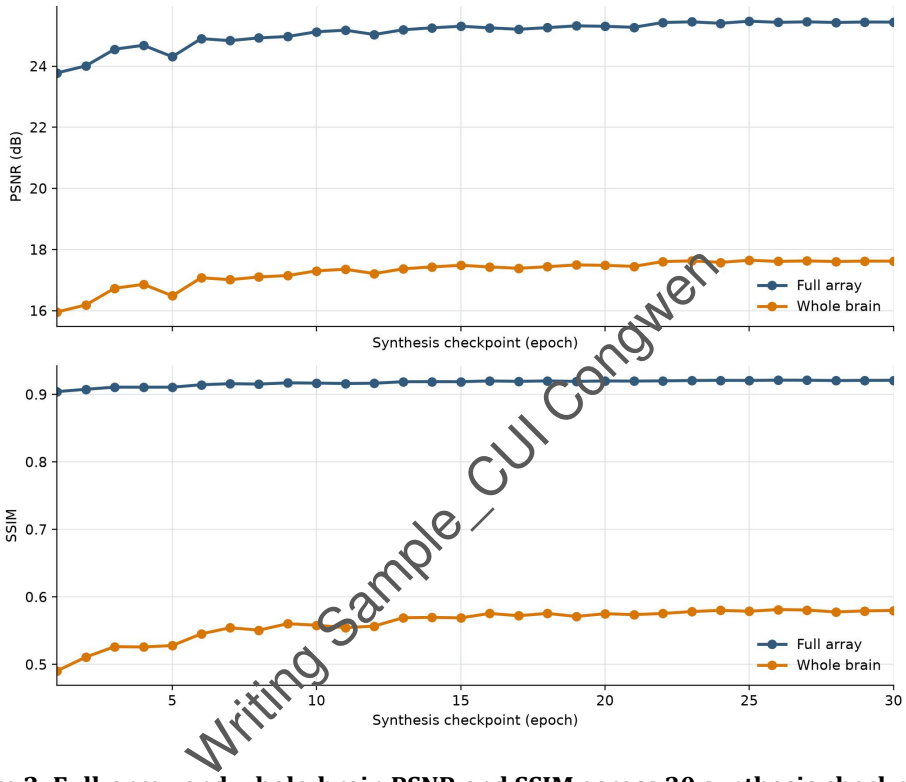
**Figure 1: BraTS 2021 modalities and their roles in synthesis and evaluation**

| Split | Patients | Purpose |
|---|---|---|
| Training | 1,001 | Train the synthesis and segmentation models |
| Validation | 125 | Select model checkpoints without using test outcomes |
| Testing | 125 | Evaluate 30 synthesis checkpoints with fixed segmentation |

**Table 1: Patient-level dataset split**

## 3.2 T1ce Synthesis Model

A 2.5D residual U-Net was trained to synthesise the centre T1ce slice from T1, T2 and FLAIR. For each modality, the model received the centre slice and its two adjacent slices, giving nine input channels. Checkpoints were saved for epochs 1 to 30. The synthesis curves improved most rapidly during the early checkpoints and then approached a plateau. Full-array measurements remained substantially higher than whole-brain measurements because the full array included the large zero-valued external background. The two scopes were therefore retained separately rather than treated as interchangeable measurements.




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**Figure 2: Full-array and whole-brain PSNR and SSIM across 30 synthesis checkpoints**

## 3.3 Downstream Segmentation Model and T1ce Control

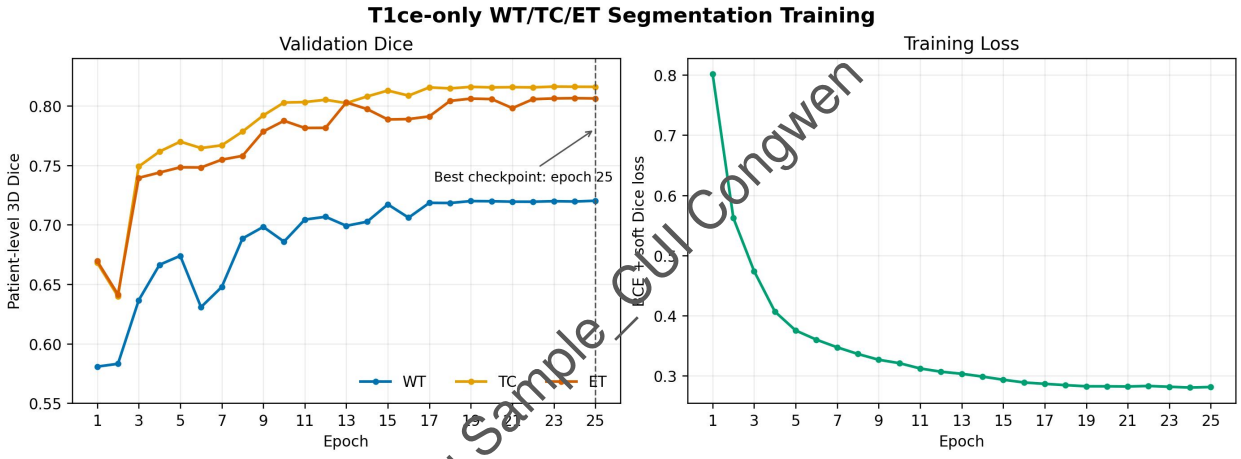
The initial downstream model used T1, T2, FLAIR and T1ce to predict WT, TC and ET. Across synthesis checkpoints, WT remained comparatively stable while TC and ET deteriorated. This pattern raised the possibility that T2 and FLAIR compensated for missing T1ce information in WT segmentation, making the four-modality model less sensitive to the synthesis checkpoint.

A T1ce ablation experiment tested this explanation while holding the patient inputs and segmentation parameters constant. Replacing T1ce with T1 reduced mean test Dice from 0.920 to 0.888 for WT, but from 0.883 to 0.219 for TC and from 0.832 to 0.044 for ET. Setting T1ce to zero produced the same regional pattern. WT therefore retained substantial information from the other modalities, whereas TC and ET depended much more strongly on contrast-enhanced information.

| Fourth input condition | WT Dice | TC Dice | ET Dice |
|---|---|---|---|
| Real T1ce | 0.920 | 0.883 | 0.832 |
| T1ce replaced by T1 | 0.888 | 0.219 | 0.044 |
| T1ce set to zero | 0.872 | 0.119 | 0.015 |

**Table 2: T1ce ablation results for the four-modality segmentation model**

The main relationship experiment therefore used a T1ce-only segmentation control with WT, TC and ET outputs. This removed compensation from T1, T2 and FLAIR and isolated the segmentation information retained in real or synthetic T1ce. Validation selected epoch 25, after which the segmentation parameters remained fixed for every test patient and synthesis checkpoint.




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**Figure 3: Validation Dice and training loss for the T1ce-only segmentation model**

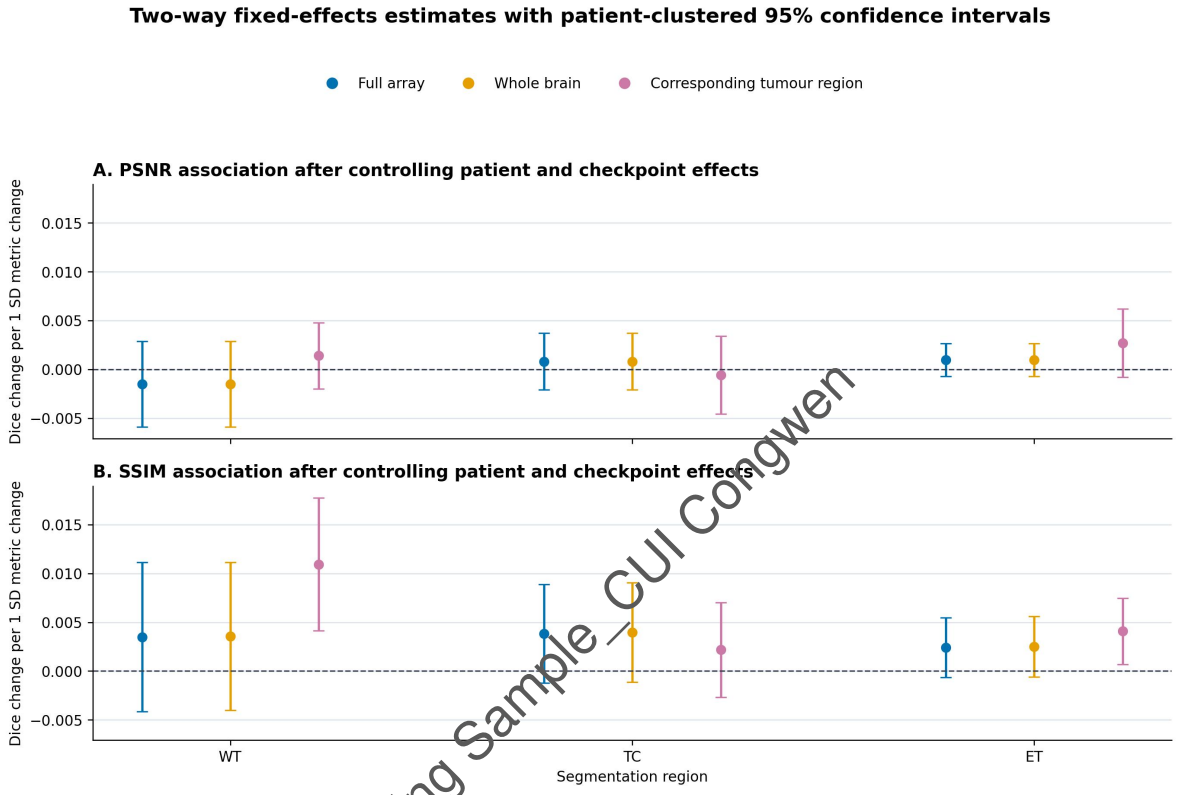
## 3.4 Checkpoint-Based Downstream Evaluation

For each of the 125 held-out test patients, the fixed epoch-25 segmentation model received the real T1ce reference and synthetic T1ce generated by each of the 30 synthesis checkpoints. This produced 3,750 patient-checkpoint observations. For every observation, PSNR and SSIM were calculated over the full volume, the whole brain and the corresponding WT, TC or ET region. Downstream performance was measured using 3D WT, TC and ET Dice.

The repeated-measures design holds the patient and segmentation model constant while the synthesis checkpoint changes. It therefore supports two complementary analyses: a checkpoint-level model of the average relationship after controlling patient and checkpoint effects, and a within-patient analysis of whether image quality and Dice move together across checkpoints.

## 3.5 Checkpoint-Level Relationship Analysis

A two-way fixed-effects regression controlled for stable differences between patients and common differences between synthesis checkpoints. Patient-clustered 95% confidence intervals accounted for repeated observations from the same patient. False-discovery-rate correction was then applied across the multiple metric-region tests.




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**Figure 4: Fixed-effects estimates for PSNR and SSIM after controlling patient and synthesis-checkpoint effects**

PSNR measured over the full volume, whole brain or local tumour region did not show a stable association with WT, TC or ET Dice. Local WT SSIM was the only image-quality predictor that remained significant after false-discovery-rate correction (q = 0.0338). Local ET SSIM had a positive estimate before correction but did not remain significant after correction (q = 0.167), while the TC confidence intervals crossed zero. The evidence therefore supports a small stable relationship for WT-local SSIM, rather than a general relationship across all metrics and tumour regions.

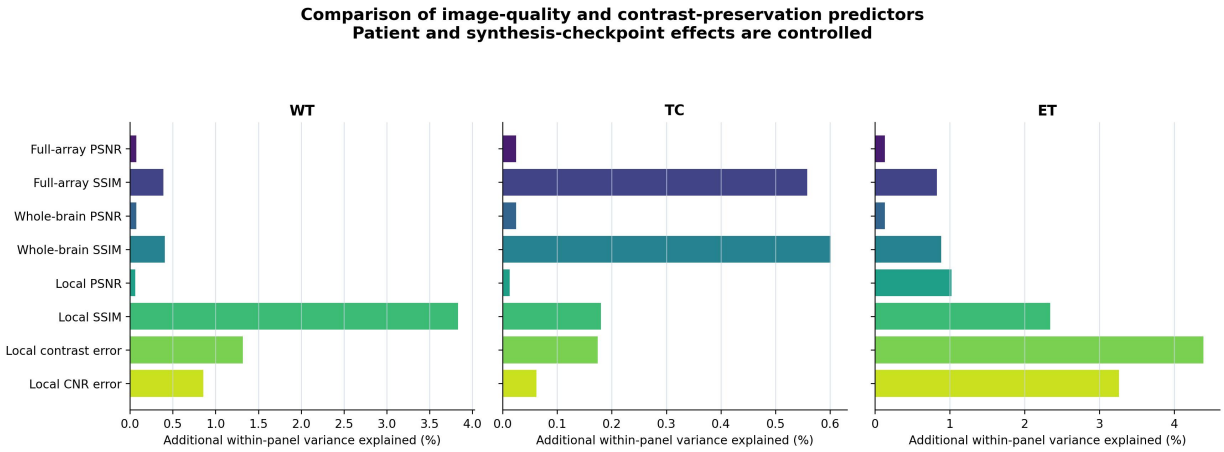



**Figure 5: Additional within-panel variance explained by image-quality and contrast-preservation predictors**

Additional R-squared values were low. WT-local SSIM explained approximately 3.8% of additional within-panel variance, while the PSNR measures contributed little additional explanatory power. Statistical significance for one predictor therefore did not imply that PSNR or SSIM explained most checkpoint-related Dice variation.

## 3.6 Patient-Level Relationship Analysis

Checkpoint-level averages can conceal patients whose relationships point in different directions. For each test patient, Spearman correlation was calculated between the 30 checkpoint values of PSNR or SSIM and the corresponding regional Dice. Correlations of at least 0.3 were described as synchronous, correlations of at most -0.3 as inverse, and intermediate values as weak or inconsistent. Cases with constant Dice were recorded as undefined. These thresholds support visual description and are not clinical decision boundaries.

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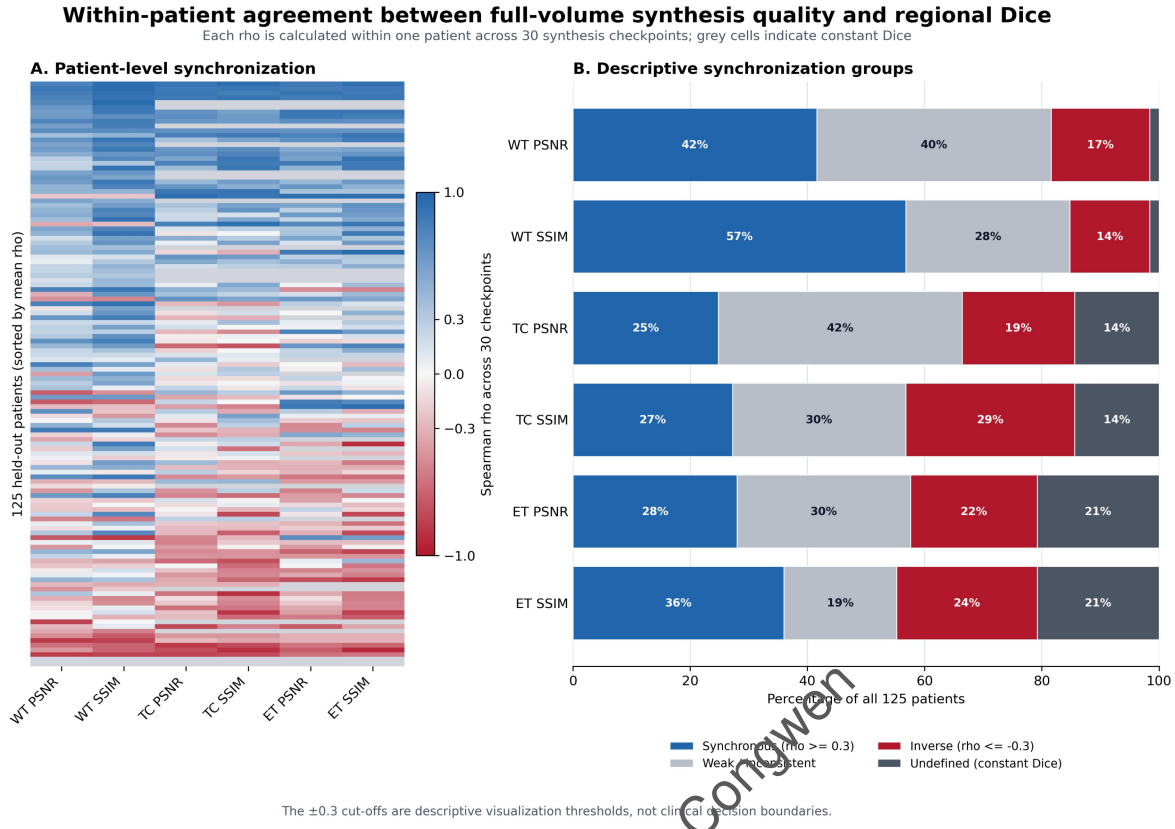
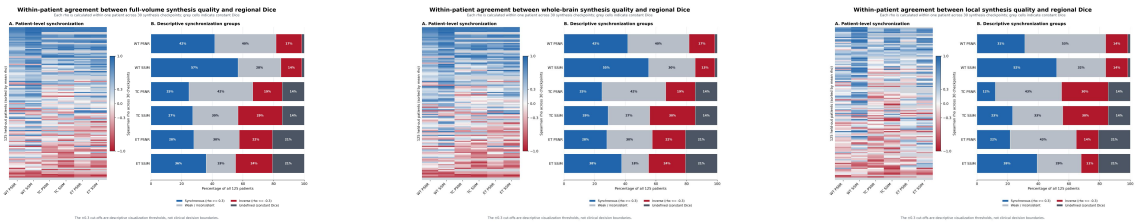



**Figure 6: Within-patient agreement between full-volume synthesis quality and regional Dice**

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WT showed the greatest synchronization, particularly for SSIM: 57% of patients were synchronous, 28% weak or inconsistent, 14% inverse and approximately 1% undefined. TC and ET showed greater patient-level variation. For TC SSIM, 27% of patients were synchronous, 30% weak, 29% inverse and 14% undefined. For ET SSIM, the corresponding proportions were 36%, 19%, 24% and 21%. The weak average relationship can therefore arise partly because positive and inverse patient-level patterns offset one another.

The same patient-level calculation was repeated for the full volume, whole brain and local tumour region. The resulting distributions changed with the evaluation mask, showing that the observed upstream-downstream relationship depends on where image quality is measured. At this stage, the analysis establishes scope-dependent change but does not yet identify its mechanism.



**Full volume** **Whole brain** **Local region**

**Figure 7: Patient-level Spearman correlations across full-volume, whole-brain and local-region evaluation scopes**

## 3.7 Completed Work Summary

The completed experiment provides a reproducible link between 30 T1ce synthesis checkpoints and fixed downstream segmentation in 125 held-out patients. The T1ce ablation clarified why the original four-modality model was relatively insensitive for WT and justified the T1ce-only control. The statistical analysis found limited evidence that conventional PSNR and SSIM consistently explain Dice changes. Patient-level results further showed that the relationship varies across patients and changes when the evaluation scope moves from the full volume to the whole brain and local tumour region.

# 4 Conclusion

The project has progressed from a small 2D pilot to a formal 2.5D checkpoint-based evaluation over 1,251 BraTS 2021 patients. The formal design uses patient-level training, validation and test splits, all 155 axial slices, 30 synthesis checkpoints and a fixed downstream segmentation model. This design separates synthesis variation from changes in the patient set or downstream model.

Ablation of T1ce in the original four-modality segmentation model showed that T2 and FLAIR could substantially compensate for WT, while TC and ET depended much more strongly on contrast-enhanced information. The main analysis therefore used a fixed T1ce-only segmentation model. In the checkpoint-level model, PSNR did not show a stable relationship with WT, TC or ET Dice, and only WT-local SSIM remained significant after multiple-comparison correction. Its additional explanatory power was limited.

Patient-level analysis showed that a weak overall association does not imply that every patient lacks an upstream-downstream relationship. Synchronous, weak, inverse and constant-Dice patterns coexisted, especially for TC and ET, and their distribution changed across full-volume, whole-brain and local-region evaluation. The next stage will characterise these scope-dependent changes and test whether tumour volume, TC/WT ratio, morphology, local contrast, texture and Dice floor effects are associated with them. Robustness and generalisability will then be evaluated using alternative models, patient subsets and an independent dataset where feasible.

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